Debeilei Wang

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Education

University of Chicago *Sept 2025 – Present*
MS in Computational and Applied Mathematics (MCAM)

- **Graduate Coursework:** Measure-Theoretic Probability I & III, Brownian Motion and Stochastic Calculus, Applied Analysis.

University of Birmingham *Sept 2021 – Jun 2025*
BSc in Applied Mathematics with Mathematics

- First-Class Honours

Jinan University *Sept 2021 – Jun 2025*
BSc in Mathematics and Applied Mathematics

- Overall Score: 90.8/100
- **Core Coursework:** General Topology (98), Differential Geometry (95), Abstract Algebra I (93), Multivariable & Vector Analysis (98), Linear Algebra (97), Real Analysis (85).

Research & Reading Projects

Directed Reading in Random Geometry and SLE *Autumn 2026 – Present*
Supervisor: Prof. Ewain Gwynne, Department of Mathematics, University of Chicago

- Study conformal invariance, Schramm-Loewner Evolution (SLE), and Gaussian Free Fields based on Lawler's *Conformally Invariant Processes in the Plane*.
- Prepared and delivered technical seminar presentations on SDE formulations, comparison theorems, and recurrence/transience transitions for Bessel processes in critical dimensions.

The Bernstein Problem for Minimal Surfaces *Sept 2024 – Apr 2025*
Undergraduate Thesis, Jinan University & University of Birmingham

- Investigated Bernstein's problem for minimal graphs in Euclidean spaces.
- Analyzed classical differential geometric proofs in dimensions $n \leq 7$ and examined the failure mechanisms for entire minimal graphs in dimensions $n \geq 9$.

Predictive Modeling on Academic Assessment (REF2021) *Jun 2024 – Oct 2024*
Undergraduate Research Bursary, School of Mathematics, University of Birmingham

- Developed machine learning pipelines in Python to evaluate and predict research assessment metrics from REF2021 data.

Teaching Experience

Jinan University *Oct 2024 – Jan 2025*
Teaching Assistant, Integer Programming and Combinatorial Optimization

Jinan University *Mar 2024 – Jun 2024*
Teaching Assistant, Statistics

Jinan University *Oct 2023 – Jan 2024*
Teaching Assistant, Multivariable and Vector Analysis

Honors & Awards

School of Mathematics Undergraduate Research Bursary, *University of Birmingham* Apr 2024

Academic Scholarship, *Jinan University* 2023, 2024, 2025